Ericsson PE Node WFD-PE-02 Whitefield POP Hardware/Environmental Fault Resolution SOP

Node: WFD-PE-02 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_7911

Description

This Standard Operating Procedure outlines the diagnosis and resolution of critical hardware/environmental fault failures on the Ericsson PE node WFD-PE-02 at Whitefield POP. The failure domain in question pertains to ASIC memory parity errors, line card crashes, and power supply redundancy loss. These issues are typically associated with the PE node's hardware, specifically the line cards and power supply units. The PE node's role is to terminate and aggregate various network interfaces, including Ethernet, SONET/SDH, and ATM. In the event of hardware failures, the PE node may exhibit erratic behavior, including but not limited to, system crashes, interface flapping, and protocol misbehavior. The diagnostic and resolution steps outlined in this SOP aim to identify and rectify the root cause of these failures, ensuring optimal network performance and reliability. The Ericsson CLI dialect is used throughout this procedure due to its specificity and relevance to the PE node's configuration and fault management features.

Pre-requisite Checks

- Verify that the Whitefield POP is accessible and that remote hands are available on-site.
- Check the status of the redundant links and ensure that they are functioning correctly.
- Review recent maintenance records and logs to identify any potential causes or contributing factors.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system status, Expected Output: System status: active, power supply: normal, line card: online, Description: This command verifies the overall system status, including power supply and line card operation. A normal output indicates that the system is functioning correctly.

Command: show line card status, Expected Output: Line card status: online, health: normal, temperature: within limits, Description: This command provides detailed information on the status of the line card, including its health and temperature. An abnormal output may indicate a line card-related issue.
```

```
Command: show power supply status, Expected Output: Power supply status: normal, redundancy: active, Description: This command verifies the status of the power supply, including its redundancy. A normal output indicates that the power supply is functioning correctly and that redundancy is active.
```

Resolution Steps

Step 1: Step: Replace the faulty line card, Command: replace line card slot 1, Description: This step involves replacing the faulty line card with a spare unit. The replacement process should be performed in accordance with the manufacturer's instructions and the site's procedures.

Step 2: Step: Replace the faulty power supply, Command: replace power supply 1, Description: This step involves replacing the faulty power supply with a spare unit. The replacement process should be performed in accordance with the manufacturer's instructions and the site's procedures.

Step 3: Step: Reset the system, Command: system reset, Description: This step involves resetting the system to its default configuration. The reset process should be performed in accordance with the manufacturer's instructions and the site's procedures.

Post-Check Verification

```
Command: show system status, Expected Output: System status: active, power supply: normal, line card: online, Description: This command verifies that the system is functioning correctly after the resolution steps.

Command: show line card status, Expected Output: Line card status: online, health: normal, temperature: within limits, Description: This command verifies that the line card is functioning correctly after the resolution steps.

Command: show power supply status, Expected Output: Power supply status: normal, redundancy: active, Description: This command verifies that the power supply is functioning correctly after the resolution steps.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: system restore original config, Description: This step involves restoring the original system configuration before the resolution steps. The rollback process should be performed in accordance with the manufacturer's instructions and the site's procedures.

- Step: Replace the replaced component, Command: replace line card slot 1 (or power supply 1), Description: This step involves replacing the replaced component with the original faulty unit. The rollback process should be performed in accordance with the manufacturer's instructions and the site's procedures.